

Cyber Shield: Defending the Network

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Cisco Virtual Internship 2025 – Cyber Security Report

Introduction

The reliance on digital infrastructure within academic institutions has grown significantly, making cybersecurity a priority. College networks connect thousands of students, faculty, and IoT devices and are often targeted by cyberattacks.

This report, prepared for the Cisco Virtual Internship Program 2025, addresses real-world cybersecurity challenges. It is divided into three parts:

- **Campus Network Security Audit** – Red-team style assessment of the current setup.
- **Hybrid Working Environment** – Design for secure access for faculty and students.
- **Web Access Policy Framework** – Restrict misuse while maintaining usability.

A risk assessment and recommendations for future improvements are also included.

Part 1: Campus Network Security Audit

The existing campus network primarily uses a star topology with multiple departmental switches connected to a core distribution switch. VLANs exist but are loosely enforced. Wireless networks for faculty, students, and guests overlap, creating risks.

Identified Security Gaps:

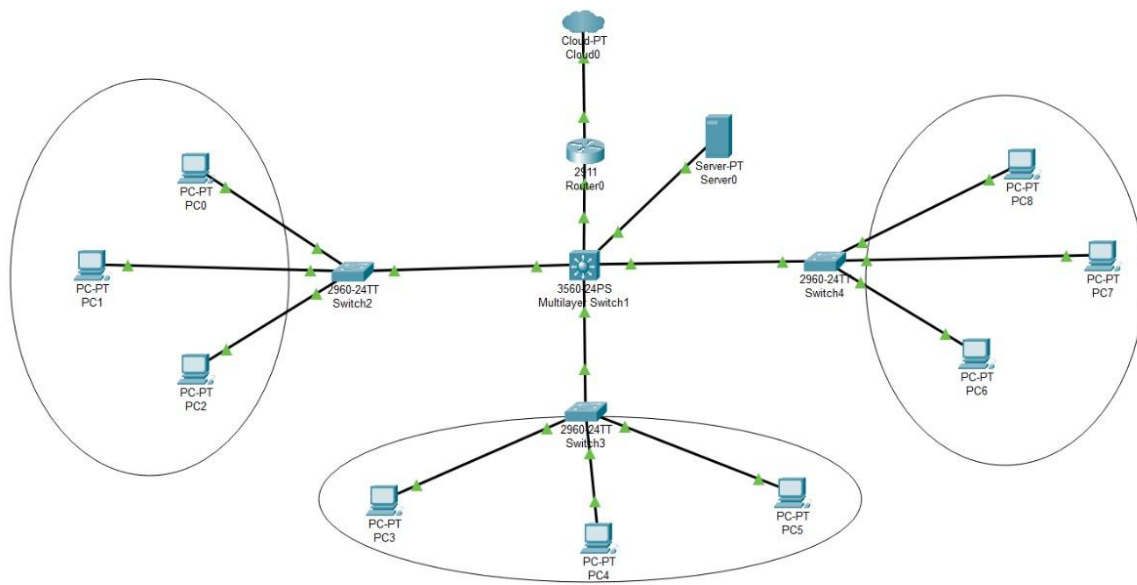
- Weak or absent ACLs between VLANs
- Guest devices able to probe internal subnets
- IoT devices (CCTV, sensors) mixed with student networks
- Management interfaces exposed to users

- No IDS/IPS or centralized monitoring

Proposed Countermeasures:

- Enforce strict VLAN segmentation (Faculty, Student, Guest, IoT, Admin)
- Apply ACLs to block lateral movement
- Deploy NAC with 802.1X authentication
- Enable DHCP snooping, ARP inspection, and port security
- Deploy IDS/IPS and centralize logging with SIEM

Diagram 1: Campus Network Topology



Sample IP Address Table:

VLAN	Subnet	IP Range	Gateway
Faculty	10.10.1.0/24	10.10.1.10–254	10.10.1.1
Student	10.10.2.0/24	10.10.2.10–254	10.10.2.1
Guest	10.10.3.0/24	10.10.3.10–254	10.10.3.1
IoT	10.10.4.0/24	10.10.4.10–254	10.10.4.1
Admin	10.10.5.0/24	10.10.5.10–254	10.10.5.1

Part 2: Hybrid Network Architecture

Faculty require secure remote access while students use personal devices on campus. Internal services must not be exposed directly to the internet.

Proposed Architecture:

- **Faculty:** Remote connection → VPN Gateway → NGFW → Core Switch → Faculty VLAN
- **Students:** Personal device → Campus Wi-Fi (NAC enforced) → NGFW → Core Switch → Student VLAN
- **NGFW:** Performs deep packet inspection and applies policies
- **Authentication:** Multi-Factor Authentication (MFA) for faculty, captive portal for students

Risks & Benefits:

- VPN ensures confidentiality but may introduce latency
 - NGFW provides strong security but requires resources
 - NAC enforces device compliance but adds complexity
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Part 3: Web Access Policy Framework

After hybrid access was enabled, students misused resources by streaming, torrenting, and bypassing filters.

Solution:

- Deploy DNS filtering for malicious and non-educational sites
 - Use Layer-7 firewall to block streaming, gaming, torrents
 - Apply role-based policies:
 - **Faculty:** Full academic & research access
 - **Students:** Education only, streaming blocked during class hours
 - **Guests:** Internet access only, no private resources
 - Log violations & generate weekly reports
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Sample Policies:

- Allow: Education, Research, Journals
- Block: Social Media, Gaming, Streaming (8 AM – 4 PM)
- Guests: Deny all RFC1918 private IP ranges

Risk Assessment & Mitigation

Risk	Impact	Countermeasure
Unauthorized Access	Data breach, misuse	NAC, VPN, MFA
Lateral Movement	Spread of malware	VLAN isolation, ACLs
IoT Exploitation	Pivot into internal network	IoT VLAN, patching, firmware updates
Misuse of Resources	Reduced productivity, bandwidth	DNS/Web filtering, policies
Network Flooding/DDoS	Downtime, service unavailability	IDS/IPS, firewall thresholds, rate-limit

Conclusion & Future Scope

This report addressed three critical cybersecurity challenges:

- Auditing the campus network to identify risks
- Designing a secure hybrid architecture for faculty and students
- Creating a policy framework to prevent misuse of resources

Future Enhancements:

- Zero Trust Network Access (ZTNA)
- AI-driven anomaly detection
- SIEM integration for visibility
- Cloud-based SASE for scalability

By implementing these, the campus network will remain secure, scalable, and resilient.

AICTE Registration profile and dashboard:



Welcome: Bhoomi

Email: 202401100058@mitaoe.ac.in

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- Profile
- Assessment Test (NEW)
- Application Management
- Private Internships
- Government Internships
- NHAI Internships
- AMRUTAM 2.0
- Problem Statements

Edit

Profile of BHOOMI NITIN RAUT

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Nationality: Indian
Physically Handicapped: NO

Documents:
Resume Transcript Government ID Other Documents OBC Caste Certificate

Social Media:

About Me

A knowledge freak, likes to explore new things, read books and newspaper, writing diary is my favorite hobby.



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Dashboard

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1 Internships Applied

0 Shortlisted

0 Interview Calls

0 Selected

0 Completed

0 Gig Applied

0 Connections Verified

0 Connections Pending for Verification

Government Internship			
ULB/Smart City Name	Internship Name	Date of Application	Status
Private Internships			
Internship Name		Date of Application	Status
CISCO VIRTUAL INTERNSHIPS PROGRAM 2025		2025-05-26 11:41:00	Applied
Rural Internship			
Rural Body Name	Internship Name	Date of Application	Status
NHAI Internship			

Thank you.....